Comparing Safety-Removal Subspaces in Aligned LLMs

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Abstract

Recent research suggests LLM safety mechanisms' relationship to general utility remains poorly understood. Several methods have been proposed to isolate or remove safety-related structure, including Difference in Means (DIM), Refusal Cone Optimization (RCO), and ActSVD. We reproduce Difference-in-Means (DIM), ActSVD, and RCO on Llama-3.1-8B-Instruct and compare them at the behavioral, geometric, and causal levels. We extend the adversarial-suffix probe of DIM from a single GCG suffix on Qwen 1.8B to WildJailbreak wrappers on Llama-3.1-8B, adding a per-method ablation cross-test in which RCO ablation strictly outperforms DIM on bare harmful requests. A benign-wrapped control shows that wrapper style itself perturbs the refusal subspace.

1 Introduction

1.1 A Contradiction in the Literature

Four recent papers make conflicting claims about whether safety is separable from utility in aligned LLMs:

Arditi et al. (2024) show that for several common chat models, a manipulation of a single dimensional subspace is enough to both induce refusal of non-harmful requests, and turn off the refusal of harmful requests. They use a procedure called **difference-of-means** to identify this subspace. This technique measures the average activations at each token position of each transformer layer, compared between a set of harmful requests and a set of harmless requests. Once these difference-of-means vectors are constructed, they score each for its ability to cause the model to refuse harmless requests and respond to harmful requests. The vector with the highest score is normed and selected as the refusal dimension vector, denoted by $\hat{r}$. This refusal dimension can then be ablated in the residual stream at inference time to bypass refusal behavior. This vector can also be used to create a jailbroken model by updating model weights according to the following formula: $W'_{\text{out}} \leftarrow W_{\text{out}} - \hat{r}\hat{r}^T W_{\text{out}}$. They find that this attack method is successful on the Qwen model family with and without the system prompt, and on the Llama model family without the system prompt, and has little effect on model coherence.

Wei et al. (2024) introduce low-rank decomposition methods designed to identify specific ranks within a weight matrix related to given LLM behaviors. Their ActSVD algorithm performs Singular Value Decomposition on the product of the model weights and input activations (WX_{in}), and yields an orthogonal projection matrix (Π).

Removing the top safety-critical ranks ActSVD identifies causes the model to completely stop rejecting unsafe prompts, and the model's utility is severely compromised. These findings suggest that safety regions in aligned models are also crucial for its general utility. To disentangle safety from utility, the authors remove safety ranks orthogonal to utility ranks using $\Delta W = (I - \Pi^u)\Pi^s W$. This yields higher attack success rate for unsafe prompts while maintaining zero-shot accuracy for utility prompts.

The fact that naively ablating safety ranks destroys utility, whereas surgically removing disentangled ranks preserves it, indicates that top safety ranks and top utility ranks heavily overlap. The necessity of this orthogonal projection matrix provides strong evidence against the hypothesis of strict linear separability between safety and utility. Ultimately, ActSVD provides rank-level evidence for superposition: safety and utility share representational capacity and are not linearly distinct.

Ponkshe et al. (2026) demonstrates fundamental mathematical limitations of linear subspace-based safety defenses, arguing that safety is not linearly separable from utility. Removing any specific safety-related subspace inherently degrades the overall performance of the model. The authors support this hypothesis through a series of empirical evaluations. Using singular value decomposition and mode subspace overlap, the study reveals that the principal directions amplifying safe behaviors also amplify useful ones, indicating that these directions do not constitute a distinctly separable safety subspace. Furthermore, attempts to mitigate harmful behaviors via orthogonal projection resulted in a proportional drop in the model's utility. The researchers also found that providing the model with helpful and harmful inputs produced highly overlapping activations. Collectively, these findings further challenge the hypothesis that safety and utility operate within linearly separable subspaces.

Wollschläger et al. (2025) generalize the identification of safety subspaces to conic regions of multiple basis refusal vectors as opposed to one refusal direction. Instead of testing pairs of harmful and harmless prompts, their methods of Refusal Direction Optimization and Refusal Cone Optimization perform gradient descent to converge on refusal vector direction(s). They leverage two properties of ideal refusal vectors in loss functions for optimization:

- Given a refusal direction r, scalar α, initial activation x_i, and revised activation $\check{x}_i = x_i + \alpha \cdot r$, refusal probability should scale with α.
- Removing the refusal direction should not affect harmless prompts while allowing response to harmful prompts.

This research finds significant jailbreaking performance gains using one refusal direction, with further gains up to a four-dimensional refusal region. Testing on Gemma-2, Llama 3, and Qwen 2.5 model families and benchmarks such as TruthfulQA, ablation of multiple refusal vectors is shown to have better attack success and lower side-effects on model performance.

Furthermore, noting that vector orthogonality does not guarantee causal independence, the research defines a notion of representational independence between vectors, in which the ablation of one direction does not impact the effects of the other. Ablating three or more representationally independent refusal vectors is found to have higher attack success than difference-of-means direction ablation. This further shows that safety and utility occupy a complex subspace within LLMs.

These claims range from clean one-dimensional separation to fundamental inseparability. If Arditi et al. (2024) are correct, a single targeted defense could block safety-removal attacks. If Ponkshe et al. (2026) are correct, safety removal inevitably degrades the model. These are materially different conclusions for alignment.

1.2 Research Questions

We investigate whether these claims can be reconciled empirically by running all three methods on the same model and asking:

1. **Cross-method agreement.** Do DIM, ActSVD, and RCO converge on the same geometric structure, despite operating at different levels (activation space vs. weight space)?
2. **Safety-utility entanglement.** How much does each method's safety direction overlap with the model's utility activation subspace? Is the *full* safety subspace entangled even if individual *selected* directions are not?
3. **Behavioral tradeoff.** When each method removes safety, how much utility does it preserve? Does geometric overlap predict behavioral utility cost?
4. **Causal independence.** Are multiple refusal directions causally independent, or does ablating one change the effect of another?
5. **Generalizing the adversarial-suffix analysis of Arditi et al. (2024).** Their §5.1 shows that a single GCG-optimized adversarial suffix suppresses the refusal direction at the EOI position on Qwen 1.8B Chat. The analysis is explicitly restricted to "a single model and a single adversarial example." We examine whether the same suppression appears under in-the-wild prompt-wrapping attacks, on a more recent and larger model (Llama-3.1-8B-Instruct), and whether the suppression is layer-localized or distributed.

1.3 Key Insight

We hypothesize that these four claims may be simultaneously correct because they describe different objects. The *full* safety subspace (e.g., the layer-wise stack of DIM mean-difference vectors) may be entangled with utility, while each method's *selection procedure* identifies a surgical direction within that entangled space with lower utility overlap. DIM selects the direction with minimum KL divergence on harmless prompts; ActSVD explicitly orthogonalizes safety against utility; RCO's loss includes a retain term penalizing harmless-prompt disruption. All three procedures implicitly optimize for low safety-utility overlap, which may explain why the resulting interventions preserve utility despite broad entanglement.

Some of the disagreement across papers may also reflect genuine differences in models, datasets, or evaluation protocols rather than contradictory truths about the same underlying phenomenon. Our controlled single-model study eliminates model variation as a confound but cannot fully resolve this ambiguity.

2 Methodology

Targeting Llama-3.1-8B-Instruct, we compare three methods for extracting safety subspaces:

Method	Operates on	Intervention
DIM	Residual stream activations	Inference-time ablation by 1 direction
ActSVD	Weight matrices and activations per linear layer	Removal of safety ranks from weight matrix
RCO	Residual stream activations	Inference-time ablation by subspace of 2 basis vectors

Table 1: The three methods operate at different levels of the model but target the same behavioral outcome (removing refusal).

2.1 Subspace Comparison

Our comparison phase addresses three questions. First, *cross-method consistency*: do DIM, ActSVD, and RCO converge on similar safety-relevant features, or does each capture a distinct aspect of the safety mechanism? Second, *safety–utility overlap*: how much does each layer's safety direction lie inside a utility activation subspace? Third, *behavioral safety-utility tradeoff*: after removing safety behavior, how much does each method degrade useful behavior?

DIM and RCO operate in activation space, while ActSVD operates in weight space. We bridge these two regimes in two complementary ways. **Weight-delta MSO**: for each layer and weight type, we compute the thin SVD of $\Delta W = W_{\text{actsvd}} - W_{\text{base}}$ and project the DIM and RCO directions onto its left singular basis. This is a **capacity** measure: the realized perturbation $\Delta W\boldsymbol{x}$ depends on the input distribution. **Activation-delta direction**: per-layer mean activation deltas computed across 64 harmless prompts provide a direct activation-space comparison between the base and ActSVD-modified models.

Mode Subspace Overlap (MSO). Following Ponkshe et al. (2026), MSO measures the geometric overlap between two subspaces. For two matrices $\boldsymbol{V}$ and $\boldsymbol{W}$, we extract their principal directions via thin SVD and select the smallest number of left singular vectors capturing an η-fraction of the energy. The MSO metric is defined as:

$$\text{MSO}(\boldsymbol{V}, \boldsymbol{W}; \eta) = \frac{\|S\|_F^2}{\min(k_V, k_W)}$$

where $S = Q_V^\top Q_W$ is the overlap matrix between the orthonormal bases. MSO ranges from 0 (orthogonal subspaces) to 1 (identical spans). We compute MSO between the DIM refusal direction and ActSVD weight-delta subspaces to assess cross-method agreement. We also compute MSO between DIM safety directions and matched utility PCA subspaces to quantify safety–utility entanglement directly. Because DIM yields a single direction while ActSVD yields multi-dimensional subspaces, cross-method MSO is bounded by the dimensionality asymmetry; we report the random baseline $\mathbb{E}[\text{overlap}] = \max(k_V, k_W)/d$ alongside each MSO value for calibration.

Representational Independence (RepInd). Following Wollschläger et al. (2025), RepInd tests whether two individual directions are *causally* related, not merely geometrically similar. Two directions $\lambda, \mu \in \mathbb{R}^d$ are representationally independent if ablating one does not change the cosine similarity profile of the other across layers:

$$\forall l \in L : \cos(\boldsymbol{x}^{(l)}, \lambda) = \cos\left(\tilde{\boldsymbol{x}}^{(l)}_{\text{abl}(\mu)}, \lambda\right)$$

and vice versa. MSO may report high geometric overlap between directions that turn out to be causally independent, or low overlap between directions that are causally entangled via nonlinear interactions across layers. In this codebase, the cones-repind implementation can train RDO directions, orthogonal directions, independent directions, and multi-vector refusal cones. Our project-owned RepInd script measures a direction's layerwise cosine-similarity profile before and after ablating another direction. Without completed optimized RCO artifacts, we derive a small cone-like basis from high-norm DIM mean-difference candidates; with trained RCO artifacts, the same script can compare DIM, RDO, RepInd, and cone basis vectors directly.

3 Experimental Settings

All experiments target Llama-3.1-8B-Instruct on a single A100 GPU. The full pipeline is reproducible from the script entry points documented in `readme.md`.

Datasets. JailbreakBench (Chao et al., 2024) provides 100 categorized harmful prompts for ASR. Harmless compliance and perplexity use Alpaca (Taori et al., 2023) generic-text perplexity uses the Pile (Gao et al., 2020). ActSVD calibration uses Alpaca-cleaned-no-safety (utility) and alignment SFT data (safety), following Wei et al. (2024). The prompt-attack probe uses HarmBench standard behaviors (Mazeika et al., 2024) for the direct-request baseline and WildJailbreak (Jiang et al., 2024) `adversarial_harmful` / `adversarial_benign` splits for the wrapped and benign-control groups (streamed with shuffle buffer; gated, requires HF token).

Method hyperparameters.

- **DIM**: 128 harmful (AdvBench split) and 128 harmless prompts; layer $l_* = 11$ selected with KL- and steerability filtering.
- **ActSVD**: 128 calibration samples; utility rank $r^u = 3950$, safety rank $r^s = 4090$, matching Wei et al. (2024)'s reported optimum (effective ΔW rank ≈ 6).
- **RCO**: 2-D cone, DIM-initialized, learning rate 1×10^{-3}, batch size 16, 1500 steps; refusal-scaling + surgical-ablation + KL-retain losses.

Evaluation harness. JBB ASR (100 prompts) is graded by substring at eval time, then regraded post-hoc by **Qwen3Guard-Gen-4B** (Zhao et al., 2025), an external response-safety classifier (1.19M-pair training set, three-tiered safe / controversial / unsafe labels), applied to every method's saved completions in a single consistent pass. Using an external moderator from a different model family removes both the cross-method confound (a method's intervention biasing its self-judgment) and the same-family bias of using the base model as judge. Harmless compliance (100 prompts), Pile/Alpaca perplexity (64 each), and TruthfulQA (64 questions, substring against `correct_answers` / `incorrect_answers`) round out the behavioral evaluation. All rates carry 1,000-sample bootstrap 95% CIs. **Two random baselines** ($\mathcal{N}(0, I)$, seed 7) are added as sanity checks: a 1-D random direction (rank-matched to DIM's ablation) and a 2-D random orthonormal subspace (rank-matched to RCO's cone). Together they distinguish direction-specific effects from rank-dependent ones. Safety-utility overlap uses PCA of 128 harmless activations at ranks $k \in \{1, 2, 4, 8, 16, 32\}$. We adopt $k = 8$ as the primary rank for the headline numbers because the full DIM mean-difference subspace's MSO-to-random-baseline ratio peaks at $k = 8$ on this model (78× at $k \in \{1, 2\}$, 71× at $k = 4$, 98× at $k = 8$, 74× at $k = 16$, 45× at $k = 32$); $k = 8$ therefore captures the strongest entanglement signal while remaining well below the 4096-dimensional residual-stream space. The full rank sweep is preserved in the supplementary results; RepInd uses 32 prompt pairs with a 3-D basis; the probe (75 prompts) adds a **layer sweep** (project at every block input) and an **ablation cross-test** (generate each prompt under base, DIM-ablated, and RCO-ablated conditions).

4 Results

4.1 Behavioral Benchmark

Method	sub-ASR	Q3G-ASR	Harmless	PPL Pile	PPL Alpaca
Base	0.15	0.00	1.00	13.93	8.60
DIM-Ablated (1-D)	1.00	0.90	1.00	14.17	8.80
ActSVD-Modified	0.80	0.77	1.00	20.16	11.65
RCO-Cone-2 (true 2-D ablation)	1.00	**0.93**	1.00	14.08	8.76
Random-Direction-7-1D	0.16	0.00	0.98	14.65	8.86
Random-Subspace-7-2D	0.14	0.00	0.99	14.61	8.78

Table 2: Behavioral benchmark results. Higher ASR = more safety removed. Higher harmless compliance = better utility preservation. Lower perplexity = better language modeling quality.

Substring ASR overestimates jailbreak success because it scores any non-refusal — including vague or hedged completions — as a jailbreak. The Qwen3Guard response-safety judge provides the discriminating metric and yields the ranking **RCO (0.93) > DIM (0.90) > ActSVD (0.77)**. The corresponding refusal-rate counts agree (RCO complies on 99/100 prompts, DIM on 98/100, ActSVD on 84/100). ActSVD's lower ASR is accompanied by substantially higher perplexity (Pile 20.16, Alpaca 11.65 vs. baseline 13.93 / 8.60), indicating that weight-space modification is both less effective at removing refusal and more disruptive to general language modeling than activation-space ablation. The two random baselines establish direction-specificity: the 1-D random direction yields ASR 0.16 / 0.00 and the rank-matched 2-D random subspace yields 0.14 / 0.00, ruling out the alternative hypothesis that RCO's gain is a mechanical consequence of ablating additional dimensions.

JBB ASR is graded by **Qwen3Guard-Gen-4B**, an external safety classifier purpose-built for response-safety classification. Jailbroken is defined as the moderator's `Unsafe` label; `Controversial` is conservatively counted as not jailbroken. TruthfulQA (64 questions, substring grading) is mostly ambiguous (78–89% across methods), so it serves only as a weak side-effect check. All rates carry 1,000-sample bootstrap 95% CIs.

4.2 Cross-Method Geometric Agreement

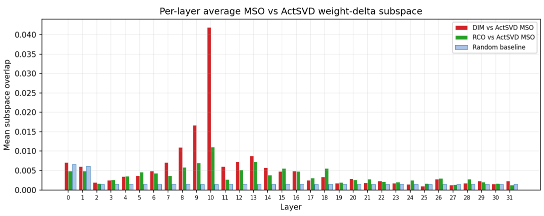

Figure 1: Cross-method geometry. Per-layer MSO to the ActSVD weight-delta column space, averaged over Q-proj, O-proj, and MLP-down. Layer 10 is the main averaged hotspot (DIM 0.042, RCO 0.011).

In Figure 1, DIM-vs-ActSVD MSO stays near the random baseline except for a layer 10 hotspot (0.042 averaged over Q-proj, O-proj, and MLP-down). RCO-vs-ActSVD shows a weaker layer 10 hotspot (0.011). Separately, the direct DIM-vs-RCO activation-space comparison gives a top principal-angle cosine of 0.505. Individual weight matrices produce larger peaks at the same layer: DIM-vs-ActSVD reaches 0.070 on MLP-down ($48\times$ random), while RCO-vs-ActSVD reaches 0.023 on attention output ($16\times$). We treat this as **exploratory rather than conclusive**: the SVD bridge characterizes the column space that ActSVD's edit **can** perturb, not its realized effect $\Delta W x$ on inputs sampled from the data distribution. The behavioral benchmark, in which all three methods elevate ASR, provides stronger evidence that they share a common refusal mechanism, even though the bridge cannot localize it precisely. The DIM-vs-RCO top principal-angle cosine is 0.505 (reducing to the ordinary cosine because DIM is one-dimensional), indicating moderate agreement in activation space; the gradient-optimized direction departs substantially from the statistical mean-difference estimate, consistent with a nontrivial difference in the loss landscape's optimum.

Safety object	MSO (rank 8)	vs. Random
Full DIM mean-diffs (averaged)	0.191	98×
DIM selected direction (layer 11)	0.078	40×
DIM selected direction (avg. over layers)	0.0165	8.4×
RCO 2-D cone (subspace MSO)	**0.0030**	**1.5×**
ActSVD activation delta (per-layer)	0.0605	31×
Random 1-D baseline	0.00195	1×
Random 2-D subspace baseline ($k_{S\{=\}}2$, $k_{U\{=\}}8$)	0.00195	1×

Table 3: Safety-utility MSO at rank 8 for each method's safety object. RCO uses the normalized 2-D subspace MSO formula $\|U^\top Q_S \frac{\|_F^2}{\min(k_S,k_U)}$ with baseline $\frac{\max(k_S,k_U)}{d} = \frac{8}{4096}$.

4.3 Safety-Utility Overlap: The Central Finding

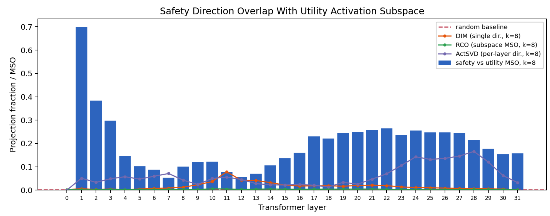

Figure 2: Per-layer safety-utility MSO at rank 8. Bars: full DIM mean-difference direction at each layer (98× random on average). Lines: each method's safety object projected onto the same utility PCA basis. RCO's 2-D cone subspace MSO (green) tracks the random baseline at every layer.

The full DIM mean-difference stack has MSO 0.191 (98× random), consistent with the claim of Ponkshe et al. (2026) that safety is broadly entangled with utility. The DIM-selected direction (40× at layer 11, 8.4× averaged) occupies a substantially less entangled region of the same space, and RCO's optimized 2-D cone has normalized subspace MSO of just 0.0030 (1.5× random) at rank 8 — essentially orthogonal to the utility PCA basis, only modestly above the normalized random baseline at the same rank. RCO ablates more dimensions than DIM yet preserves perplexity at least as well, indicating that the dimensions it ablates contain less utility-relevant content. The gap between the full mean-difference stack and the selected direction partly reflects an object-level asymmetry: the full measure averages all per-layer candidates, including high-overlap early layers, while the selected direction is KL-filtered for intervention. The method-level comparison (DIM single direction at 8.4× vs. RCO 2-D cone at 1.5×) uses the same rank-$k = 8$ utility basis and dimension-aware normalization. We emphasize that PCA-MSO is not a sufficient predictor of utility cost; the perplexity column of Table 2 serves as the definitive behavioral test.

4.4 Representational Independence

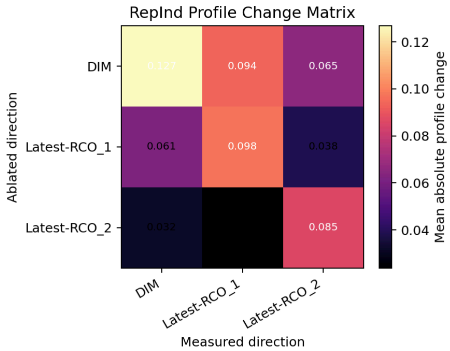

Figure 3: RepInd profile-change matrix. Rows: ablated direction. Columns: measured direction. Lower off-diagonal values indicate greater representational independence.

The RepInd analysis reveals an asymmetric causal structure. Ablating DIM changes the first RCO cone basis vector's per-layer cosine profile substantially (mean $|\Delta| = 0.094$, max $|\Delta| = 0.218$), while ablating that same basis vector causes a smaller change to DIM's profile (mean $|\Delta| = 0.061$, max $|\Delta| = 0.280$). The second RCO basis is still more independent: ablating DIM changes its profile by mean $|\Delta| = 0.065$, but ablating it changes DIM by only mean $|\Delta| = 0.032$. This asymmetry indicates that DIM captures a dominant refusal component that causally influences other directions, but the reverse is weaker — consistent with Wollschläger et al. (2025)'s finding that multiple representationally-independent refusal mediators coexist, with a hierarchy of causal influence rather than fully orthogonal channels.

4.5 Prompt-Based Jailbreaks vs. the Refusal Direction

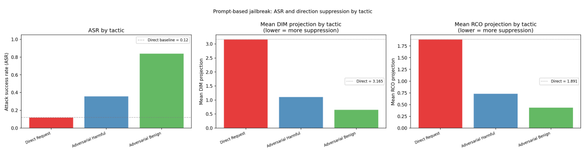

Figure 4: ASR (left) and mean DIM/RCO projection (right) per prompt group.

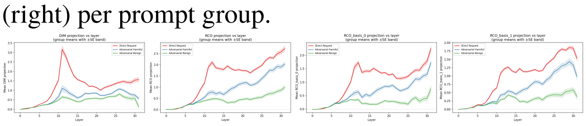

Figure 5: **Layer sweep:** per-layer mean DIM/RCO projection per group, ±SE bands.

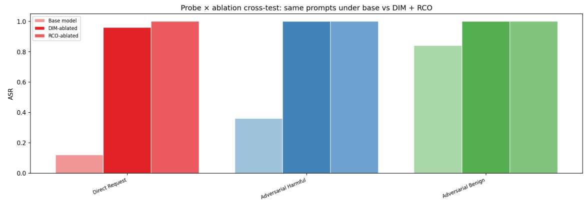

Figure 6: **Ablation cross-test:** same probe prompts under base vs each ablated model. Both DIM (1-D direction ablation) and RCO (2-D cone subspace ablation) are run as separate conditions on the **same** prompts. If RCO's cone covers refusal directions DIM misses, RCO-ablated ASR should be at least as high as DIM-ablated ASR on every group.

The probe metric for RCO is the L2 norm of the EOI activation's projection into the optimized 2-D cone subspace, $\text{proj}_{\text{RCO}} = \left\|B^\top \boldsymbol{h}\right\|_2$, which is the natural multi-dimensional analog of DIM's signed scalar $\text{proj}_{\text{DIM}} = \boldsymbol{h} \cdot \hat{\boldsymbol{r}}$. We also record the two per-basis signed scalars $\text{proj}_{\text{RCO,basis 0}}$ and $\text{proj}_{\text{RCO,basis 1}}$ separately for diagnostic value.

The probe yields a coherent picture. Direct harmful requests exhibit low base ASR (0.12) and high projection onto both refusal subspaces ($\text{proj}_{\text{DIM}} = 3.17$, $\text{proj}_{\text{RCO}} = 1.89$ at layer 11; per-basis 1.55 and 1.08). Adversarial-harmful wrapping increases ASR to 0.36 and reduces all projections in tandem ($\text{proj}_{\text{DIM}} = 1.11$, $\text{proj}_{\text{RCO}} = 0.73$; per-basis 0.60 and 0.42), generalizing the §5.1 pattern of Arditi et al. (2024) from a single GCG suffix on Qwen 1.8B to a population of WildJailbreak attacks on Llama-3.1-8B. The benign-wrapped control suppresses the projections further ($\text{proj}_{\text{DIM}} = 0.66$, $\text{proj}_{\text{RCO}} = 0.44$; per-basis 0.33 and 0.29) while producing high compliance (ASR 0.84). The reduction in projection is therefore not uniquely associated with harmful jailbreaks; the wrapper style itself perturbs the refusal subspace.

The ablation cross-test applies DIM and RCO ablations separately to the same probe prompts. DIM ablation lifts direct-request ASR from 0.12 to 0.96, while RCO 2-D cone ablation lifts it to 1.00 — a strict improvement on the bare-harmful subset where DIM leaves four residual refusals out of 25 prompts. On the adversarial groups both ablations reach 1.00, indicating that the cone's additional coverage manifests precisely where DIM's single direction is weakest. This is consistent with the behavioral advantage of RCO in Table 2 and with the geometric picture: the cone covers refusal-mediating directions that the DIM-selected vector misses while maintaining utility entanglement at random-baseline levels. The layer sweep localizes the DIM gap sharply around the selected layer $l_* = 11$ for direct vs. adversarial-harmful prompts; the RCO subspace norm grows into later layers, suggesting it captures a related but more distributed refusal geometry.

4.6 Self-Consistency

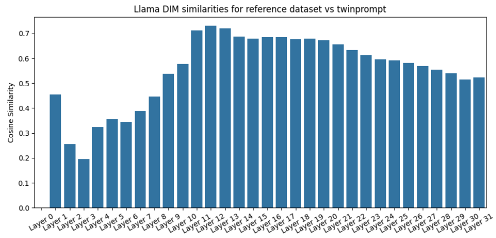

Figure 7: Difference-in-Means Self-Consistency

The figure above shows the cosine similarity of the difference-in-means candidate vectors when trained on the reference dataset vs when trained on TwinPrompt (Krauß et al., 2025). The fact that the similarity is non-uniform in part reflects that these two datasets are simply different; we don't expect all the DIM candidates all to reflect safety. However, if the DIM process truly identifies a single universal safety vector for the model, we should expect perfect agreement on that subspace. The reference dataset identifies the vector at layer 11 as the safety vector, and the graph above shows that the cosine similarity at layer 11 is 0.7126. This is not perfect agreement but fairly close. The highest agreement is at layer 12 with a cosine similarity of 0.7303.

4.7 Layer-wise Activation Divergence: ActSVD vs. Diff-in-Means

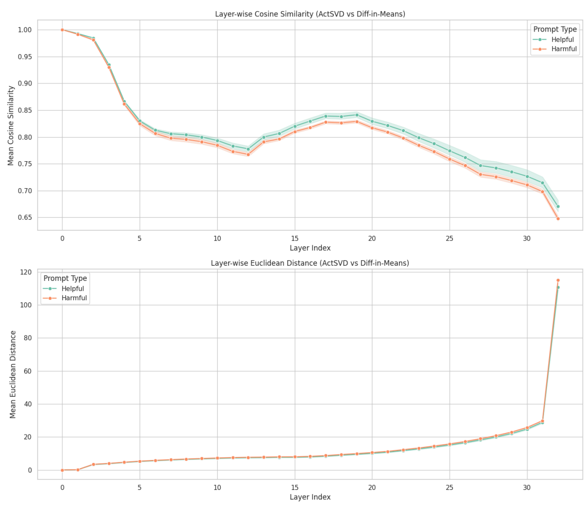

Figure 8: Cosine similarity and Euclidean distance between ActSVD and DIM jailbroken model activations

This comparison examines how the two editing methods differ from each other across all 33 layers (embedding layer + 32 transformer blocks). In both cosine similarity and Euclidean distance, we observe that the two modified models produce clearly different internal representations. Harmful prompts are slightly more different than helpful, but not by a large margin. These findings are counter to what we would expect if these methods are truly only affecting safety, which would imply that each model would act similarly to the base model when prompted on helpful prompts.

4.8 Layer-wise Activation Divergence: Base vs. Modified Models

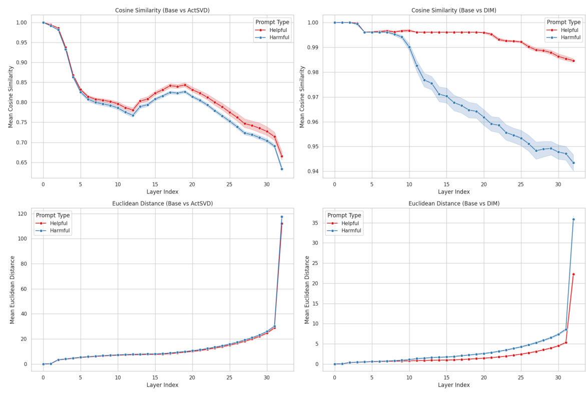

Figure 9: Cosine similarity and Euclidean distance between ActSVD and DIM jailbroken model activations and the Base model. Note the different Y axis scaling between the graphs

In the second set of comparisons (Base vs. ActSVD and Base vs. DIM), we see that ActSVD changes the base model much more than DIM, especially for helpful prompts. If a method is only affecting a safety subspace, we would expect the Base and the modified model to be very similar on helpful prompts while being different on Harmful prompts.

For ActSVD, this is not the case at all. We see that both harmful and helpful prompts are significantly affected by the modification, indicating it is also affecting utility along with safety.

For DIM, helpful prompts *are* affected much less than harmful ones, but they are still significantly affected again indicating it is also affecting utility along with safety.

5 Discussion

Our work in this paper has shown some promising new leads. Through our comparison of the overlap of the subspaces generated by difference-in-means (DIM), ActSVD, and Refusal Cone Optimization

(RCO), we have shown the existence of multiple seperate safety subspaces. This shows that Arditi et al.'s (2024) titular claim that "Refusal in Language Models Is Mediated by a Single Direction" is inaccurate. On one hand, this is promising for LLM safety as it shows that safety mechanisms are distributed throughout the model. One the other hand, the fact that any one of these methods can be successful without touching the other methods safety subspaces shows that these safety mechanisms are brittle. They do not function as failsafes for one another.

As far as implications for the field, the fact that methods like we have described here work so well should be concerning. If our goal as a field is to release models that cannot be trivially modified to behave in unsafe ways, the success of a simple method like difference-in-means shows we still have a long way to go. From a safety perspective, it is desirable for safety and utility to be entangled. Models where safety and utility are seperable are inherently vulnerable to these kinds of attacks. Therefore future work in jailbreak resistance research should specifically aim to entangle safety and utility, such that any attempt to make the model respond to unsafe requests makes the model useless.

5.1 Limitations

Our work has some limitations.

- **Only one model family:** All of our experiments use Llama-3.1-8B-Instruct. Our results may not generalize to other families of models such as Qwen or Deepseek.
- **Only one model size**: We used an 8B parameter model for all our experiments. It may be the case that larger models have more deeply intertwined safety subspaces.
- **PCA utility $\neq$ causal utility.** The utility subspace is defined by variance, not causal contribution; low MSO does not prove low utility damage. Behavioral perplexity is the definitive test.
- **RepInd uses the RCO cone basis as the second pair of candidates**, not independent vectors from RepInd's own loss; the asymmetry is evidence about the cone vs. DIM, not a fully general independence test.

6 Conclusion

In this paper we have replicated the results of multiple state-of-the-art jailbreak methods. We have compared the safety subspaces generated by each of these methods and shown that they do not entirely overlap. This shows that while safety may be distributed throughout multiple subspaces, this does not lead to a more robust safety mechanism.

7 Member Contributions

All five authors contributed to writing and research-design discussion. Effort (totaling 100%):

7.1 Evan Scamehorn (20%)

Integrated reference paper methodology implementations. Led project planning and git maintenance.

7.2 Adam Venton (20%)

Implemented activation comparison experiment. Wrote experiment analysis

7.3 Calvin Kosmatka (20%)

Conducted literature survey. Implemented and analyzed self-consistency experiment for DIM.

7.4 Kyle Sha (20%)

Implemented Jupyter notebooks and pipeline scripts. Implemented and analyzed prompt-attack probe (incl. layer sweep + ablation cross-test) and WildJailbreak. Wrote discussion.

7.5 Zeke Mackay (20%)

Wrote RepInd analysis, conducted RCO training, and wrote asymmetry interpretation. Conducted dataset analysis.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in Language Models Is Mediated by a Single Direction. In *Advances in Neural Information Processing Systems*, pages 136037–136083.

Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwag, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramèr, Hamed Hassani, and Eric Wong. 2024. JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models. In *Advances in Neural Information Processing Systems*.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. 2020. The Pile: An 800GB Dataset of Diverse Text for Language Modeling. *arXiv preprint arXiv:2101.00027*.

Liwei Jiang, Kavel Rao, Seungju Han, Allyson Ettinger, Faeze Brahman, Sachin Kumar, Niloofar Mireshghallah, Ximing Lu, Maarten Sap, Yejin Choi, and Nouha Dziri. 2024. WildTeaming at Scale: From In-the-Wild Jailbreaks to (Adversarially) Safer Language Models. In *Advances in Neural Information Processing Systems*.

Torsten Krauß, Hamid Dashtbani, and Alexandra Dmitrienko. 2025. TwinBreak: Jailbreaking LLM Security Alignments based on Twin Prompts. arXiv: 2506.07596 [cs.LG].

Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. 2024. HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal. In *Proceedings of the 41st International Conference on Machine Learning*, pages 35181–35224.

Kaustubh Ponkshe, Shaan Shah, Raghav Singhal, and Praneeth Vepakomma. 2026. Safety Subspaces are Not Linearly Distinct: A Fine-Tuning Case Study. In *The Fourteenth International Conference on Learning Representations*. arXiv: 2505.14185 [cs.LG].

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Alpaca: A Strong, Replicable Instruction-Following Model. Stanford CRFM blog.

Boyi Wei, Kaixuan Huang, Yangsibo Huang, Tinghao Xie, Xiangyu Qi, Mengzhou Xia, Prateek Mittal, Mengdi Wang, and Peter Henderson. 2024. Assessing the Brittleness of Safety Alignment via Pruning and Low-Rank Modifications. In *Proceedings of the 41st International Conference on Machine Learning*, pages 52588–52610.

Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. 2025. The Geometry of Refusal in Large Language Models: Concept Cones and Representational Independence. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 66945–66970.

Haiquan Zhao, Chenhan Yuan, Fei Huang, Xiaomeng Hu, Yichang Zhang, An Yang, Bowen Yu, Dayiheng Liu, Jingren Zhou, Junyang Lin, and others. 2025. Qwen3Guard Technical Report. *arXiv preprint arXiv:2510.14276*.